

Tableau

A Practical Basics Guide for Analysts coming from Power BI & Excel

Concepts • Architecture • Diagrams • Examples • Interview Q&A

1. What is Tableau?

Tableau is a data visualization and business intelligence platform, now owned by Salesforce, used to connect to data, build interactive visualizations, and share them as dashboards. Its core idea is **"see and understand your data"** through drag-and-drop visual analysis, rather than writing chart code — conceptually the same goal as Power BI, but with its own vocabulary, engine, and philosophy.

Coming from Power BI: Almost everything you already know maps across, just with different names — a Power BI "report" is a Tableau "workbook," a "visual" is a "worksheet" or "sheet," and a Power BI "dataset/semantic model" is closest to a Tableau "data source." The mental model of connect → shape → visualize → publish → share is identical.

1.1 The Tableau product family

Product	What it's for	Power BI equivalent
Tableau Desktop	Author workbooks, connect to data, build visualizations (installed application)	Power BI Desktop
Tableau Prep	Visual, no-code data cleaning and shaping before analysis	Power Query / Dataflows
Tableau Server	Self-hosted platform to publish, schedule, and share workbooks within an organisation	Power BI Report Server
Tableau Cloud	Microsoft-hosted (SaaS) version of Tableau Server, no infrastructure to manage	Power BI Service
Tableau Public	Free, public version for publishing workbooks visible to anyone on the web	(no direct Power BI equivalent)

2. Tableau Architecture — From Data to Dashboard



A **Workbook** is the whole file: it holds all worksheets, dashboards and stories together

Fig. 1 — The building blocks inside one Tableau workbook, smallest to largest.

2.1 Vocabulary map (Power BI ↔ Tableau)

Power BI term	Tableau term
Report (.pbix file)	Workbook (.twbx file)
Visual / chart	Worksheet (or "sheet")
Page (of a report)	Dashboard
Dataset / semantic model	Data source
DAX measure	Calculated field
Power BI Service workspace	Tableau Server/Cloud project

3. Connecting to Data — Live vs. Extract

Just like Power BI's Import vs. DirectQuery choice, Tableau asks you to choose a connection type the moment you connect to data.



Tableau's extract engine is called **Hyper** — comparable to Power BI's VertiPaq in-memory engine

Fig. 2 — Live connection mirrors Power BI's DirectQuery; Extract mirrors Power BI's Import mode.

Tableau	Power BI equivalent	Behaviour
Live connection	DirectQuery	Queries the source in real time; no local copy
Extract (.hyper file)	Import	Snapshot stored locally/on server, refreshed on a schedule

4. Dimensions, Measures & Discrete vs. Continuous

Every field Tableau pulls from a data source is automatically classified in two ways, and understanding both is fundamental to building any visualization.

Classification	Meaning	Example	Default colour in Tableau's UI
Dimension	Qualitative, categorical data — used to slice/group the view	Region, Product Name, Order Date (as a category)	Blue pill
Measure	Quantitative, numeric data — usually aggregated (sum, avg, count)	Sales, Profit, Quantity	Green pill
Discrete	Distinct, separate values — creates headers/labels in the view	Category names, individual dates as headers	Blue pill
Continuous	An unbroken range of values — creates axes in the view	Sales amount, a date treated as a timeline	Green pill

Key idea: In Tableau, "discrete vs. continuous" is really about how a field is used (headers vs. axis), while "dimension vs. measure" is about the field's role (grouping vs. aggregating) — the two ideas overlap but are not identical, unlike in Power BI where this distinction is far less visible to the report author.

5. Calculated Fields — Tableau's Equivalent of DAX

A **calculated field** is a formula you write once and reuse anywhere in a workbook, similar to a DAX measure or calculated column in Power BI. Tableau's formula language is simpler and closer to Excel functions than DAX.

```
IF [Sales] > 1000 THEN "High" ELSE "Low" END
SUM([Profit]) / SUM([Sales])
```

5.1 Level of Detail (LOD) Expressions

LOD expressions let you compute an aggregation at a different granularity than what's currently shown in the view — Tableau's answer to the kind of context-manipulation Power BI does with `CALCULATE` in DAX.

LOD type	Syntax	Meaning
FIXED	{FIXED [Region] : SUM([Sales])}	Aggregate at exactly this level, ignoring the view's other dimensions
INCLUDE	{INCLUDE [Customer] : SUM([Sales])}	Aggregate at a finer level than the view, then roll up
EXCLUDE	{EXCLUDE [Customer] : SUM([Sales])}	Remove a dimension from the view's level of detail before aggregating

6. Filters — And the Order They Run In

Tableau applies different filter types in a fixed sequence, which explains some "why doesn't my number match" surprises — very similar to debugging filter context in Power BI.



Fig. 3 — Tableau's filter execution order — a Context Filter runs before (and can dramatically speed up) filters placed after it.

Why Context Filters matter: Setting a filter "to context" turns it into a temporary table computed first, so every filter after it (and any LOD FIXED calculation) operates only on that reduced data — useful both for performance and for controlling what a FIXED LOD expression actually sees.

7. Combining Data — Joins, Blends & Relationships

Method	How it works	Best for
Join	Combines tables at the row level (like SQL joins) into one logical table, before aggregation	Tables from the same connection at the same granularity
Relationship (Tableau's newer data model)	A flexible, non-destructive link between tables; Tableau decides the right join automatically per query, and each table keeps its own granularity	Most modern data-source modelling — the default and recommended approach today
Blend	Combines data from two separate data sources at an aggregated level, using a linking field	Mixing data across two entirely different connections (e.g., a spreadsheet and a database)

Relationships vs. Power BI: Tableau's newer "Relationships" concept (introduced in 2020) is directly analogous to how Power BI models relationships between tables in a star schema — each table keeps its native granularity, and Tableau resolves the correct join behind the scenes per visualization.

8. Publishing & Sharing — Tableau Server / Cloud

Once a workbook is built in Tableau Desktop, it's published to **Tableau Server** (self-hosted) or **Tableau Cloud** (Salesforce-hosted SaaS) so others can view and interact with it in a browser, similar to publishing a Power BI report to a workspace in the Power BI Service.

Concept	Tableau	Power BI equivalent
Container for content	Project	Workspace
Scheduled data refresh	Extract refresh schedule	Scheduled refresh
Row-level security	User Filters / Row-Level Security via published data sources	Row-Level Security (RLS) roles
Access roles	Site Administrator, Site Role (Creator/Explorer/Viewer)	Admin/Member/Contributor/Viewer

9. Quick Glossary

Term	Meaning
Workbook	The overall Tableau file (.twbx), containing all worksheets, dashboards and stories
Worksheet	A single chart/view built from one or more fields
Dashboard	A combination of multiple worksheets arranged together, often interactive
Story	A sequence of worksheets/dashboards presented like a slideshow to tell a narrative
Pill	The small draggable box representing a field once placed on a shelf in the view
Shelf	The drop zones in the workbook (Rows, Columns, Marks, Filters) where you place pills
Hyper	Tableau's in-memory data engine used for extracts, built for very fast query performance
LOD expression	A calculation that fixes, includes, or excludes dimensions to control its level of aggregation

10. Top 10 Interview Questions on Tableau

1 What is Tableau, and how does its architecture differ from Power BI at a high level?

Tableau is a data visualization and BI platform for connecting to data and building interactive dashboards, similar in goal to Power BI but with its own terminology and engine. Structurally, a Tableau workbook holds worksheets (individual charts), which are combined into dashboards, which can be sequenced into stories — roughly mapping to Power BI's visuals-on-a-page-in-a-report structure, but with an extra "story" layer for narrative presentations.

2 What is the difference between a Live connection and an Extract in Tableau?

A Live connection sends every query straight to the source database in real time, so the view is always current but performance depends on the source system — this is the equivalent of Power BI's DirectQuery. An Extract takes a compressed, saved snapshot of the data using Tableau's Hyper engine, giving fast performance similar to Power BI's Import mode, refreshed on a schedule rather than queried live.

3 Explain the difference between a dimension and a measure in Tableau.

A dimension is qualitative, categorical data used to slice or group a view — such as Region or Product Name — shown as a blue pill by default. A measure is quantitative, numeric data that gets aggregated (sum, average, count) — such as Sales or Profit — shown as a green pill by default. This distinction drives how Tableau automatically builds axes and headers when fields are dragged into the view.

4 What are LOD (Level of Detail) expressions, and when would you use FIXED versus INCLUDE or EXCLUDE?

LOD expressions compute an aggregation at a different level of granularity than the current view. FIXED computes at exactly the specified dimension(s), ignoring whatever else is in the view — useful for things like "total sales per customer regardless of what filters are applied." INCLUDE adds a finer dimension into the calculation before rolling back up to the view's level, while EXCLUDE removes a dimension from the view's level of detail before aggregating — both are useful for calculations like "percent of a customer's total" nested inside a broader view.

5 What is the order of operations for filters in Tableau?

Tableau applies filters in a fixed sequence: Extract Filters first, then Data Source Filters, then Context Filters, then Dimension Filters, and finally Measure Filters last. Context Filters are especially important because they create a temporary, filtered dataset that everything after them (including FIXED LOD calculations) operates on — which explains many "my numbers don't match" situations if the order isn't understood.

6 What's the difference between joining tables, blending data, and using relationships in Tableau?

A join combines tables at the row level from the same connection, similar to a SQL join, and requires matching granularity. A blend combines data from two separate data sources at an aggregated level using a common linking field — useful when mixing entirely different connections. Relationships, Tableau's newer and now-default data modelling approach, flexibly link tables without forcing a single granularity, letting Tableau resolve the correct join automatically per visualization — conceptually similar to how Power BI models relationships in a star schema.

7 What is a calculated field, and how does it compare to a DAX measure in Power BI?

A calculated field is a reusable formula defined once in a Tableau workbook and used across multiple worksheets, similar in purpose to a DAX measure in Power BI. The key difference is the formula language: Tableau's calculation language is closer to Excel-style functions and is generally considered simpler to learn than DAX, though it's less powerful for advanced context-manipulation scenarios that DAX's CALCULATE function handles.

8 What is the difference between a discrete and a continuous field in Tableau?

A discrete field produces distinct, separate values and renders as headers or labels in the view (e.g., individual category names). A continuous field represents an unbroken range of values and renders as an axis (e.g., a sales amount or a date treated as a timeline). This is a separate concept from dimension/measure — a date field, for instance, can be used either discretely (as year/month headers) or continuously (as a true timeline axis).

9 How does publishing and sharing work in Tableau Server or Tableau Cloud?

A workbook built in Tableau Desktop is published to a Project on Tableau Server (self-hosted) or Tableau Cloud (Salesforce-hosted SaaS), where it can be viewed and interacted with in a browser, refreshed on a schedule if using extracts, and secured with row-level security via published data sources or user filters — conceptually parallel to publishing a Power BI report into a workspace in the Power BI Service.

10 If you're used to Power BI, what are the biggest conceptual differences to watch for when learning Tableau?

The vocabulary differs (workbook vs. report, worksheet vs. visual, calculated field vs. DAX measure), but the bigger shifts are: Tableau's drag-and-drop "shelves" (Rows/Columns/Marks) build charts more visually and iteratively than Power BI's visual gallery; the explicit discrete/continuous distinction has no direct Power BI equivalent; and LOD expressions, while conceptually similar to DAX's CALCULATE, use a very different FIXED/INCLUDE/EXCLUDE syntax that takes some practice to internalise.